

TECHNICAL PROCESS SPECIFICATION

Revised 22-Galaxy Toy Object Optimisation

Displayed-frame data flow, convergence-controlled optimisation and decision pathway for SEP and MTOjects



Purpose

Define the auditable path from the visually revised clean calibration sample to reproducible parameter optimisation within the galaxy area of interest, followed by selection of frozen SEP and MTOjects configurations and application to the full 182-galaxy population.

Configuration documented: 29 August 2026

Environment: Windows host + Ubuntu 24.04 under WSL2 | Python virtual environment: /root/venvs/pythonscripts

Document map

This document distinguishes data artefacts from programs and identifies the control points that preserve comparability between SEP and MTOjects.

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1. End-to-end batch architecture



Key design rule

SEP and MTOBjects do not receive separately randomised toys. Both methods read the same materialised delta image, truth mask, truth labels and toy metadata from one checksum-protected manifest.

The process separates calibration from deployment. The 22 galaxies are used to tune and compare parameters. The current continuation pauses after optimisation, held-out evaluation, convergence assessment and parameter-stability reporting so the evidence can be reviewed. The next planned stage freezes one configuration per method and applies it without retraining to the 182-galaxy geometry manifest.

Primary stages

1. Selection: retain only the 22 fields classified Clean in the corrected wide, negative, galaxy-centred review.
2. Injection: create cross-validation and winner-selection toy sets with different deterministic seeds.
3. Optimisation: run leave-one-galaxy-out Optuna studies independently for SEP and MTOBjects.
4. Selection: score every fold winner on the same independent 22-galaxy winner-selection set.
5. Decision gate: compare objective convergence, near-optimal parameter stability, fold transfer and runtime before choosing a frozen configuration.
6. Full-population application: use the chosen SEP and MTOBjects configurations on all 182 galaxies and generate the usual method-specific and combined diagnostics.

2. Input sample: the revised 22 clean galaxies

The clean sample is an explicit data file rather than a filename convention or an inferred rank cutoff. It is derived from the latest categorical re-review and contains exactly the galaxies labelled Clean.

Clean galaxies 1-8	Clean galaxies 9-16	Clean galaxies 17-22
1. NGC4765	9. NGC4639	17. NGC4450
2. NGC3486	10. NGC4981	18. NGC7531
3. NGC4405	11. NGC3684	19. NGC3359
4. IC1954	12. NGC0289	20. NGC3627
5. NGC4579	13. NGC4102	21. NGC2903
6. NGC3681	14. NGC0986	22. NGC1097
7. PGC013821	15. NGC3227	
8. NGC1367	16. NGC0578	

Input	Role	Validation
clean_galaxies_revised22.txt	Authoritative calibration membership	22 unique names; exact match to current Clean decisions
s4g_image_geometry_manifest.csv	182-galaxy image paths and geometry	Required centre, PA, inclination, bar size and pixel scale
*.phot.1.fits	Original 3.6 micrometre science images	SHA-256 recorded for each injected case
candidate_union_rereview_decisions.csv	Selection provenance	22 Clean, 5 Ambiguous, 51 Polluted at completion

3. Toy creation and insertion



Two injection sets are generated for every clean galaxy. The cross_validation set is used inside fold training; the winner_selection set is independently seeded and is used to choose among the 22 fold winners. Each set contains five toys per galaxy, giving 110 toys per set and 220 materialised toys in total.

Property	Implementation
Types	Star 50%; compact cluster 20%; small galaxy 30%
Brightness	Uniform peak amplitude from 6 to 30 robust background sigma
Nominal size	Stars/clusters: 2-10 px FWHM; galaxies: 5-22 px FWHM
Analysis area	Finite pixels in the displayed, galaxy-centred, deprojected square used for visual Clean/Ambiguous/Polluted classification
Placement	Inside that displayed galaxy area of interest, away from its boundary and non-overlapping with earlier toys; unused parts of a larger FITS mosaic are excluded
Truth	Per-toy label image plus union truth mask, dilated by one pixel
Reproducibility	Global set seed plus CRC32-derived per-galaxy seed
Materialisation	Compressed NPZ containing delta, truth_mask and truth_labels; hashes stored in JSON

Adaptive placement fallback

The generator first attempts the requested five toys at normal size within the displayed deprojected area of interest. If a dense or small region cannot accommodate them, it retries at FWHM scales 0.85 and 0.70, then retries progressively fewer toys at original size. Requested count, actual count, scale, analysis-region pixel count and fallback flag are recorded per galaxy. In the current displayed-frame manifest all 44 cases (22 galaxies x 2 injection sets) accepted five toys at normal size; no fallback was used.

Correction from earlier runs

Earlier optimisation results used mask fractions and/or toy placement relative to a larger FITS frame. Those results are not comparable with the visual classifications and are treated as superseded. The current metric version is paired-toy-metrics-displayed-frame-v2.

Why materialise?

A failed or altered random draw cannot make one method appear better. SEP and MTOBjects consume identical pixels and identical truth definitions.

4. SEP masking program flow



SEP processing is provided by `sep_processing.sep_products()`. It builds a background model, subtracts it, convolves with the selected filter kernel and calls `sep.extract`. The segmentation is filtered by maximum area, elongation and central exclusion; retained components are dilated to form the final mask.

Optimised SEP parameter	Search space
<code>detect_thresh</code>	0.6-2.0 sigma
<code>minarea</code>	5-35 pixels
<code>deblend_nthresh</code>	16, 32 or 64
<code>deblend_cont</code>	0.001-0.03, logarithmic
<code>back_size</code>	32, 64, 128 or 256
<code>filter_size</code>	1, 3, 5, 7 or 9
<code>dilation_radius</code>	1-6 pixels
<code>max_area</code>	20-8000 pixels
<code>max_elongation</code>	1.5-30

The cleaned image replaces masked pixels with the median of finite unmasked pixels for diagnostic display. Optimisation is driven by the mask-to-truth comparison rather than the cosmetic appearance of that replacement.

5. MTOBjects masking program flow



MTOBjects processing is provided by `mtobjects_spike_gate_processing.mtobjects_products()`. It preprocesses the finite image, constructs an OriginalMaxTree, floods and filters the tree, relabels significant components, applies the common geometric filters and dilates the retained segmentation.

Optimised MTOBjects parameter	Search space
<code>move_factor</code>	0.0-1.0

Optimised MTOBJECTS parameter	Search space
min_distance	0.0-1.0
gaussian_fwhm	0.0-5.0
bg_variance	10 ⁻⁴ to 10 ⁴ ; calibrated/log search in this run
minarea	1-40 pixels
dilation_radius	1-6 pixels
max_area	20-3000 pixels
max_elongation	2.0-15.0

Concurrency constraint

MTOBJECTS workers are separate processes, not threads, because the MTOBJECTS library changes its working directory while loading C libraries. Linux fork allows prepared arrays to be shared copy-on-write.

6. Mask scoring and objective functions

Each method is scored against the materialised truth mask within the same displayed deprojected area of interest. The case evaluator records pixel recall, precision, F-score, mean per-toy recall, whole-toy detection rate, displayed-frame masked fraction, displayed-frame false-positive fraction and segment count.

Component	SEP weighting	MTOBJECTS weighting
Pixel recall	0.45	Included through F-score
Pixel F-score	0.20	0.45
Mean per-toy recall	0.25	0.35
Whole-toy detection rate	0.20	0.20
Data loss	0.35 x mean masked fraction	0.35 x mean masked fraction
False positives	0.05 x capped false-positive fraction	0.05 x capped false-positive fraction
Hard controls	Large penalty above 15% maximum masked fraction	Same cap plus minimum recovery feasibility gates

The optimiser minimises objective. For a feasible trial under the displayed-frame mask cap, objective is the negative of the composite score. Cap violations receive a large positive penalty. MTOBJECTS additionally rejects trials that produce no incremental mask pixels or fail its minimum toy-detection and mean-toy-recall criteria.

7. Optuna and 22-fold optimisation



Setting	Value
Fold design	Leave one galaxy out: 22 folds; 21 train + 1 held out
Trial budget	Minimum 40; maximum 80 total completed trials per fold

Setting	Value
Sampler start	8 initial TPE startup points; subsequent trials are guided
Objective early stop	After at least 40 trials, stop after 20 completed trials without meaningful improvement
Meaningful improvement	Relative tolerance 0.1% of objective scale with absolute floor 1e-5
Parallel workers	4
Methods	SEP completes first; MTOjects then runs using the same injection manifest
Study storage	Persistent SQLite studies below /root/clean22-displayed-frame-5toy-optuna-studies
Winner selection	Each fold winner is rescored on the common independent 22-galaxy winner_selection set; minimum objective wins
Outputs	Best JSON, fold summaries, candidate CSV, held-out detail CSV and configuration JSON

A held-out fold is never used to fit the candidate parameters for that fold. Its role is to estimate transfer to an unseen galaxy. The final selection adds a second protection: all 22 candidate winners are compared on the same independently injected dataset.

Objective and model-parameter stability

Objective convergence alone does not demonstrate that the masking model is well identified. Several materially different parameter combinations can occupy a near-equal objective plateau. A separate read-only audit therefore examines the elite trials (normally the best 20%, bounded to 5-20 trials) without changing Optuna sampling or the early-stop decision.

Diagnostic	Interpretation
Normalised numerical IQR	Spread of each numerical parameter relative to its declared search range; IQR above 0.15 is flagged
Categorical dominance	Most common elite value; dominance below 70% is flagged
Near-optimal clusters	Connected groups in normalised parameter space; multiple groups indicate alternative near-optimal states
Strong correlations	Absolute Spearman correlation of at least 0.75 identifies compensating parameter relationships
Cross-fold stability	Median/IQR or dominant value of each winning parameter across the available folds
Combined classification	Objective stable/continuing crossed with parameters stable/multimodal

Selection implication

A broad stable near-optimal region is preferable to an isolated winning trial. Objective-stable but multimodal folds are retained and reported; parameter instability is diagnostic evidence, not currently a hard reason to run indefinitely.

8. Orchestration, monitoring and restart behaviour

Program	Responsibility
generate_paired_toy_manifest.py	Create and checksum both injection sets; apply adaptive placement
run_clean22_full_cross_validation.py	Run SEP 22-fold CV, then MTOjects 22-fold CV

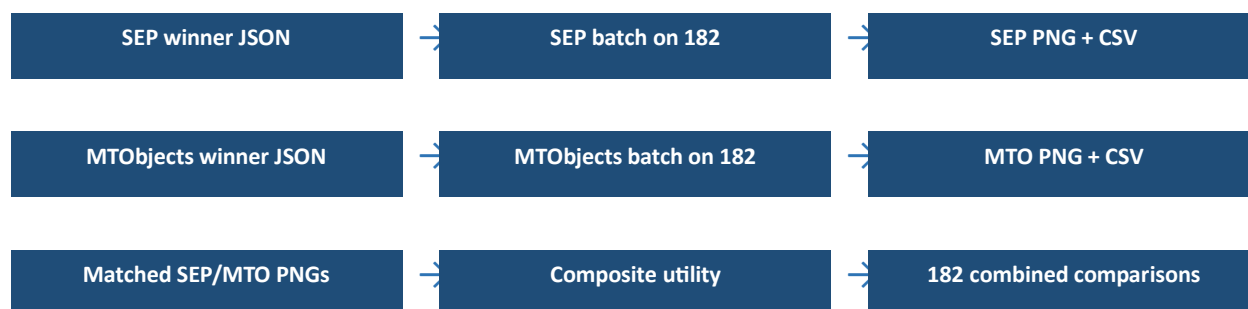
Program	Responsibility
cross_validate_toy_objects_SEP.py	Manage SEP folds, held-out scoring and final winner
cross_validate_toy_objects_MTOjects.py	Manage MTOjects folds, feasibility checks and final winner
optimise_toy_objects_SEP.py	Optuna objective and worker evaluation for one SEP fold
optimise_toy_objects_MTOjects.py	Optuna objective and process-worker evaluation for one MTOjects fold
watch_clean22_displayed_frame_5toy_80trial.sh	Resume studies, recover freezes and enforce the optimisation-only 80-trial continuation
monitor_clean22_80trial_convergence.ps1	Visible progress plus objective/parameter stability classifications
audit_optuna_parameter_stability.py	Read every SQLite study; quantify elite dispersion, clustering, correlations and cross-fold stability
watch_optuna_parameter_stability.sh	Refresh parameter-stability outputs every two minutes until completion

Completed trials remain in persistent SQLite storage and are reused. An interrupted RUNNING trial is marked failed and replaced rather than counted as complete. Candidate selection searches best-result JSON files across original and continuation directories, preventing a resumed fold from selecting only its newest trials. The watchdog regards 30 minutes without an update as a freeze and restarts the affected process chain cleanly.

Runtime benchmarking

Every trial summary records elapsed_seconds, worker count, Python version, platform and parameter values. Each cross-validation root also receives fold_timing.csv with start/end times, wall seconds, training and held-out counts, trials before/after, new-trial count, completion reason, hostname, logical CPU count and worker count. This supports later local 4/8/12-worker tests and like-for-like Azure benchmarking.

9. Next stage: application to the full 182-galaxy population



Current execution boundary

The active convergence-controlled continuation performs optimisation and validation only. This is a one-off pause for review, not removal of the 182-galaxy application from the methodology.

The next stage begins after both methods have been reported and the objective/parameter-stability evidence has been reviewed. Parameters will then be frozen and applied to all 182 galaxies; there will be no further Optuna training on that population. The geometry manifest supplies each image path and galaxy geometry.

Production entry point	Input	Output
batch_toy_objects_SEP.py	SEP winner JSON + 182-row manifest	One SEP diagnostic PNG per galaxy and sep_optimised_apply_summary.csv
batch_toy_objects_MTOjects.py	MTOjects winner JSON + MTOjects library + manifest	One MTOjects diagnostic PNG per galaxy and mtobjects_optimised_apply_summary.csv
combine_toy_method_pngs.py	Matched SEP and MTOjects PNG directories	One side-by-side comparison PNG per galaxy

- Expected production directories: SEP, MTOjects and Combined under the clean-22 all182 application root.
- The future production design will use the agreed displayed-frame toy configuration; adaptive placement metadata must be retained if a field requires fewer toys.
- Per-galaxy errors are logged and resumable. Completion requires 182 successful unique galaxy rows or explicit resolution of any exception.
- The combined view is generated after both method-specific PNG sets are complete.

10. Data lineage and audit trail

Artefact	What it proves
candidate_union_rereview_decisions.csv	Human categorical basis for the 22 clean fields
clean_galaxies_revised22.txt	Exact calibration membership consumed by the batch
paired_toy_injection_manifest.json + SHA-256	Seeds, source hashes, payload hashes, toy properties, fold membership and fallback use
payloads/cross_validation/*.npz	Exact injected deltas and truth used during fold optimisation
payloads/winner_selection/*.npz	Independent common comparison set used to choose the final winner
cross_validation_config.json	Run arguments, parameter bounds and fold membership
*optimisation_summary.csv	Trial-level objectives, metrics, status and parameters
fold_timing.csv	Per-fold wall time, new and cumulative trial counts, worker count and machine/runtime metadata
optuna_parameter_stability.csv/.json	Live fold classifications, elite dispersion, clusters, correlations and unstable parameters
optuna_parameter_stability_across_folds.csv	Cross-fold spread of the currently winning parameter values
*toy_cross_validation_best.json	Selected production parameters and winning-fold provenance
*optimised_apply_summary.csv	Per-galaxy 182-population completion and errors

Interpretation boundary

The 22 galaxies are low-contamination calibration fields, not empty images and not perfect truth. Artificial toys provide the known detection truth; the clean fields reduce interference from unknown foreground sources.

11. Acceptance checks

- Selection file contains exactly 22 unique names and matches the latest Clean decisions.
- Both injection sets contain all 22 galaxies, valid hashes and a recorded actual toy count.
- SEP and MTOObjects read the same injection manifest and named injection sets.
- Every fold completes at least 40 trials and then either satisfies the 20-trial objective-convergence rule or reaches 80 completed trials.
- Parameter-stability outputs cover all available SEP and MTOObjects SQLite studies and identify multimodal/unstable elite parameters.
- Per-trial elapsed_seconds and per-fold timing records are present for runtime benchmarking.
- Cross-validation candidate CSV contains 22 fold winners per method.
- Winner JSON identifies the selected fold and independent 22-galaxy metrics.
- The one-off review gate is completed and one SEP and one MTOObjects configuration are explicitly frozen before full-population processing.
- The next-stage production gate requires complete SEP, MTOObjects and Combined outputs for all 182 galaxies or documented exceptions.

Appendix A. Program-to-data dependency map

Producer	Data artefact	Consumer
Interactive catalogue reviewer	candidate_union_rereview_decisions.csv	Clean-list preparation
Clean-list preparation	clean_galaxies_revised22.txt	Manifest generator and CV drivers
Manifest generator	paired manifest + NPZ payloads	SEP and MTOObjects optimisers/CV
SEP CV + stability audit	SEP candidates, winner and stability evidence	Parameter decision; later SEP all-182 batch
MTOObjects CV + stability audit	MTO candidates, winner and stability evidence	Parameter decision; later MTOObjects all-182 batch
SEP and MTO batches	Method PNGs + per-galaxy summaries	Composite generator and comparison report
Composite generator	Combined PNG set	Visual review and final study documentation

Appendix B. Principal storage locations

Location	Contents
Foreground Masking/Optimisation	Clean list, manifest generator, optimisers, CV drivers and monitors
Foreground Masking/Shared	SEP and MTOObjects masking implementations and common display logic
Foreground Masking/Batch tools	All-galaxy method runners and composite orchestration
.../clean22_displayed_frame_5toy_optimisation/paired_injections	Immutable displayed-AOI clean-22 injection manifest and payloads
.../clean22_displayed_frame_5toy_optimisation/SEP_cross_validation	SEP folds, timing, trial summaries and winner

Location	Contents
.../clean22_displayed_frame_5toy_optimisation/MTOBJECTS_cross_validation	MTOBJECTS folds, timing, trial summaries and winner
.../clean22_displayed_frame_5toy_optimisation/optuna_parameter_stability*	Live within-fold and cross-fold stability evidence
/root/clean22-displayed-frame-5toy-optuna-studies	Persistent SEP and MTOBJECTS SQLite studies
Future all182 application root	Deferred SEP, MTOBJECTS and Combined production outputs